CS466 Final Project Report: Privacy-Gated Memory for Multimodal Agents

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GitHub Repository

Abstract

As AI agents increasingly persist multimodal user interactions in long-term vector memory, the risk of sensitive personally identifiable information (PII) leaking through both text and image channels grows substantially. Existing privacy tools treat PII filtering as a text-only preprocessing step, leaving visual document content entirely unguarded. We present a privacy-gated multimodal agent memory system that intercepts sensitive content before CLIP embedding through a four-component detection layer—NER-based text redaction, zero-shot visual object detection (OWL-ViT), deterministic face detection (Haar cascades), and ephemeral OCR—and applies a stage-aware policy (ALLOW / MASK / ABSTRACT) based on the fraction of detected sensitive image area. We evaluate the system with a controlled "detective-game" experiment in which GPT-4o attempts to recover private attributes from preprocessed identity document images under three conditions. Our full pipeline reduces sensitive attribute recovery by 76% relative to an unguarded baseline (attribute recovery: $0.63 \rightarrow 0.15$; recall: $0.654 \rightarrow 0.160$), while preserving identity-level task accuracy at 100%. Text-only filtering reduces attribute recovery by only 6%, confirming that visual masking is a necessary component of multimodal privacy protection.

Introduction

Modern AI agents increasingly persist user interactions as episodic memory, storing and indexing both text and image inputs in vector databases for long-term retrieval. When memory systems operate without privacy controls, sensitive personally identifiable information (PII)—names, identification numbers, financial details, medical records—can be durably embedded and later recovered through similarity search or direct agent reasoning.

Existing approaches treat privacy as a one-time text preprocessing step, e.g., Microsoft Presidio. This approach fails for multimodal inputs because vision-capable models such as GPT-4o can directly read text visible inside images. Redacting a user's text prompt while forwarding unmasked document images provides no effective protection—the model simply reads the suppressed content from the image. Also, the existing mainstream LLM protects users' data in the redaction stage, rejecting queries that directly ask for sensitive information. However, as we will show in the experiments part, this is not effective in protecting sensitive information from deceptive requests.

We address this gap with a privacy gate that operates at the sensing and memory-write stages of a multimodal agent pipeline. The gate detects sensitive regions in both text and images *before* any content is vectorized, and applies one of three actions based on the proportion of sensitive content detected. The key architectural principle is that privacy enforcement must happen before entering

LLM: once sensitive content is encoded into a dense vector, it cannot be meaningfully extracted or removed.

Our main contributions are:

- A multimodal sensing gate combining NER-based text redaction, zero-shot visual detection, deterministic face detection, and ephemeral OCR into a unified detection pipeline.
- A stage-aware policy engine that determines whether to allow, mask, or fully suppress image content based on detected sensitive area coverage.
- A controlled evaluation protocol that measures sensitive attribute recovery under three preprocessing conditions, demonstrating a 76% reduction when our full pipeline is applied.

Related Work/Tool boxes

Text PII detection. NER-based redaction tools such as Microsoft Presidio [1] operate over plain text, identifying and replacing entity spans such as names, addresses, and financial identifiers. BERT-based models fine-tuned on annotated corpora reliably detect named entities but frequently miss structured numeric identifiers—credit card numbers, national ID numbers—that appear without surrounding prose context. Our system augments NER with regular expressions targeting these patterns explicitly.

Visual privacy detection. Prior work on visual PII largely focuses on faces and license plates using supervised detectors trained on fixed category sets. OWL-ViT relaxes this constraint through zero-shot grounding of natural-language category descriptions in image features, enabling detection of categories such as "passport" or "medical prescription" without domain-specific fine-tuning.

Shared embedding spaces and cross-modal leakage. CLIP maps text and images into a shared vector space, enabling cross-modal retrieval. This shared space introduces a novel privacy risk: a benign text query can retrieve a sensitive image if their embeddings are proximate in the shared space. We mitigate this by ensuring only post-gate (masked or abstracted) content is passed to the CLIP encoder, so the stored embedding reflects the sanitized representation rather than the original sensitive content.

Problem Formulation

We model a multimodal agent memory event as a paired input $X = (x_{\text{text}}, , x_{\text{image}})$, where x_{text} denotes a text string and x_{image} denotes an image. Since user inputs may contain only text or only an image, the pipeline is designed to handle missing modalities and process the available input accordingly.The pipeline processes X through a sequence of stages

$$s \in \{\textsc{Sensing},\ \textsc{Detecting},\ \textsc{Overwriting},\ \textsc{Output}\}$$

The detection function $\textsc{Detect}(X) \to E$ produces a set of sensitive entities E, where each entity carries a type, modality, and spatial extent. A policy function maps findings and stage to an action:

$$\pi(E,\, s) \;\longrightarrow\; a \;\in\; \{\textsc{Allow},\ \textsc{Mask},\ \textsc{Abstract},\ \textsc{Block}\}$$

As for the baseline, the naive pipeline applies no policy and passes all content unchanged through every stage.

A critical limitation of text-only filtering in this formulation is that it addresses only the textual component of x_{text}. When x_{image} contains embedded text (e.g., a printed ID number on an identity card), a downstream vision-capable model recovers that information regardless of whether x_{text} was redacted. Effective privacy therefore requires π to operate over both modalities jointly.

Methodology

Our pipeline (`src/privacy_pipeline.py`) consists of three sequential stages: a sensing gate, a policy engine, and a memory write gate. Figure 1 illustrates the architecture.

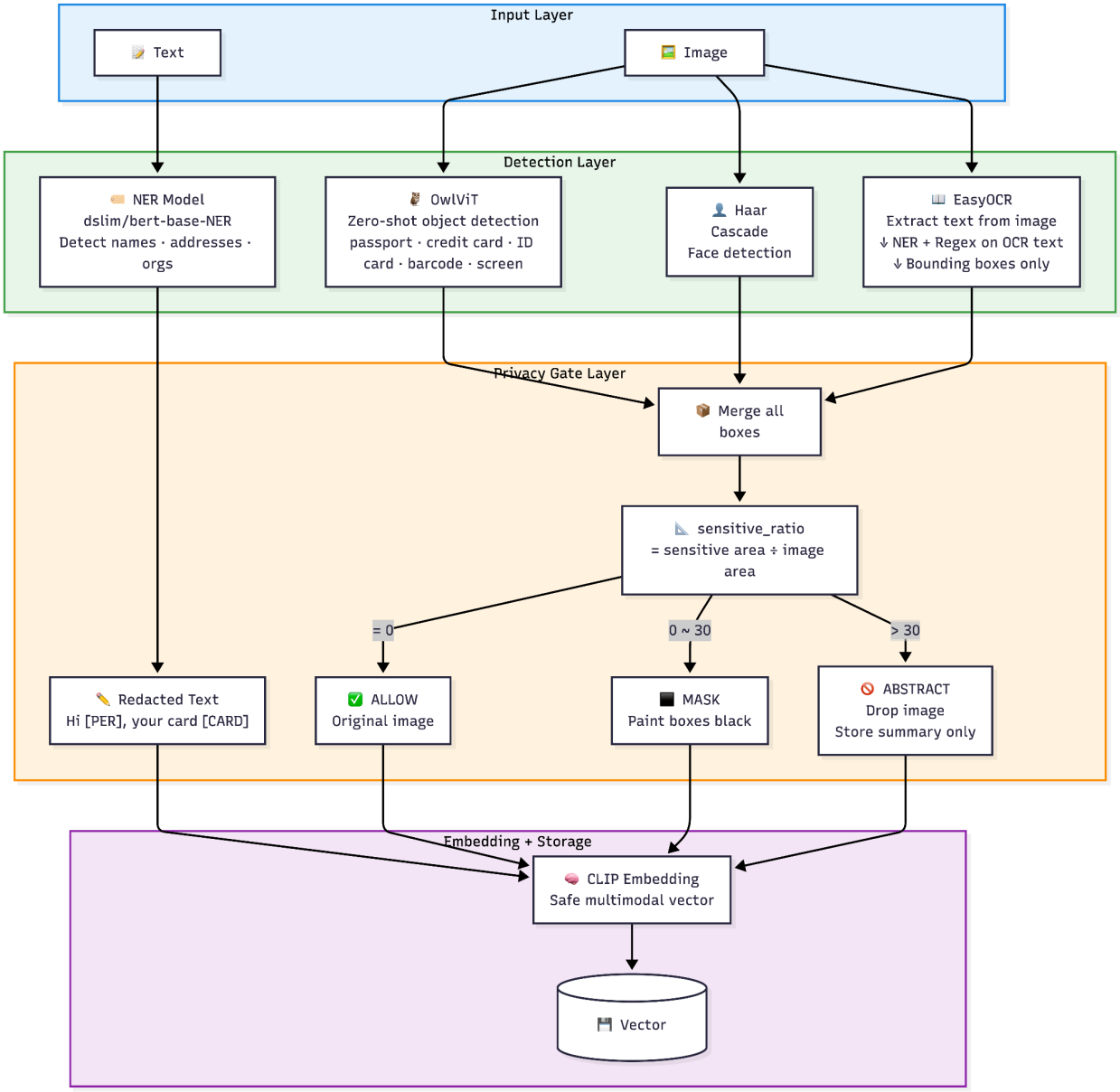

Figure 1: Pipeline Flowchart

Sensing Gate: Detection Layer

The sensing gate runs on raw input before any storage or embedding. It has four components:

1. **Text NER.** Input text is passed through `dslim/bert-base-NER`, a BERT model fine-tuned on CoNLL-2003, which detects person names, locations, and organizations. Detected spans are replaced with typed placeholders (e.g., `[PER]`, `[LOC]`).
2. **Regex patterns.** Three supplementary patterns catch structured identifiers that NER models frequently miss in short, decontextualized strings: credit and debit card numbers (14–19 digit sequences), US Social Security Numbers, and alphanumeric government IDs (up to two letter prefix followed by eight or more digits).
3. **Zero-shot visual detection.** OWL-ViT (`google/owlvit-base-patch32`) is queried with privacy-sensitive category descriptions including *passport*, *credit card*, *ID card*, *medical prescription*, *laptop screen*, *student card*, and *barcode*. A confidence threshold of 0.10 is used to maximize recall, accepting false positives in preference to missed detections.
4. **Haar cascade face detection.** OpenCV frontal and profile Haar classifiers provide deterministic face coverage. This step is necessary because OWL-ViT zero-shot queries for "face" are unreliable in real photographs; Haar cascades provide stable, CPU-efficient, device-local detection without requiring model downloads.
5. **Ephemeral OCR.** EasyOCR extracts text strings from image regions. Each string is immediately checked by the NER model and regex patterns; if sensitive content is found, the bounding box is retained but the text string is deleted. OCR-extracted text is never persisted beyond the detection step.

Policy Engine and Memory Write Gate

After detection, the policy engine computes a sensitive area ratio:

$$r = \frac{\sum_{b \in B} \text{area}(b)}{W \times H}$$

where B is the set of merged bounding boxes and $W \times H$ is the total image area. The policy action is determined by:

- $r = 0$: **ALLOW** — no sensitive regions detected; the original image passes unchanged.
- $0 < r < 0.30$: **MASK** — sensitive bounding boxes are painted solid black using PIL; the surrounding content is preserved at original resolution.
- $r \geq 0.30$: **ABSTRACT** — more than 30% of the image is sensitive; the image is dropped entirely and replaced with a fixed text summary. This prevents residual document-structure information from leaking through a partially masked representation.

For text, all NER spans and regex matches are replaced with entity-type placeholders before this stage.

Safe Embedding

Post-gate content is embedded using `openai/clip-vit-base-patch32`. A masked image produces a CLIP embedding of its masked form; an abstracted image produces an embedding of its text summary. The vector store therefore never encodes the original sensitive content, preventing cross-modal retrieval from reconstructing suppressed information.

Data

We evaluate the pipeline using synthetic document images drawn from the MultiPriv dataset [2], a synthetic PII corpus developed by CyberChangAn that provides document images annotated with privacy-sensitive fields in English and Chinese. We selected English-language samples and manually constructed three complete identity profiles, each representing a distinct fictional individual.å

Each individual is supported by multiple synthetic evidence images spanning five document categories: identity cards, bank and credit cards, flight boarding passes, medical examination reports, and customs payment receipts. Table 1 lists the nine ground-truth sensitive attribute fields used for evaluation scoring.

Field	**Privacy Category**	**Primary Source Document**
Name	Direct identifier	Identity card
ID Number	Direct identifier	Identity card
Address	Direct identifier	Identity card
Bank Card Number	Financial	Credit / bank card
Flight Number	Travel	Boarding pass
License Plate	Vehicle	Customs receipt
Diagnosis	Medical	Medical report
Salary	Financial	Ground truth only
Relationship	Social	Ground truth only

Table 1: Ground-truth sensitive attribute fields used in the detective-game evaluation.

The three profiles represent subjects from different countries (United States, Germany, China), providing variation in address formats, flight routes, ID number structures, and document styles. Evaluation was limited to English-language content in the NER path; regex patterns provided supplementary coverage for structured numeric fields regardless of surrounding language.

Experiments and Evaluation

We evaluate the system from two perspectives: a *detection-mode ablation* that isolates the contribution of each pipeline component, and an *agent-level leakage experiment* that measures how much sensitive information a downstream LLM can extract after different preprocessing conditions are applied.

Detection Mode Ablation

To assess the contribution of each pipeline component, we compare four detection modes that progressively combine filtering mechanisms:

Mode	Text Filter (NER + Regex)	Visual Detection (OwlViT + Haar)	Ephemeral OCR
A — Text-Only	✓		
B — Visual-Only		✓	
C — Visual + OCR		✓	✓
D — Full Pipeline	✓	✓	✓

Table 2: Ablation modes comparing privacy detection components.

Mode A demonstrates the limitation of text-only filtering: sensitive text provided directly in the prompt is redacted, but PII embedded within images is ignored entirely. Mode B shows that visual detection identifies sensitive objects (identity cards, faces) but misses privacy-sensitive text contained inside otherwise unremarkable document regions. Mode C adds ephemeral OCR to recover image-embedded text, and Mode D combines all three for maximum coverage. In addition to privacy detection, image utility is estimated as the cosine similarity between CLIP embeddings of the original and post-gate images; a higher score indicates that semantic content is better preserved after masking.

Baselines

Raw baseline. Images are sent to GPT-4o unmodified with no text or visual preprocessing. This establishes the upper bound on attribute leakage—what a vision-capable model can extract when no privacy protection is applied.

Text-only baseline. Text context is labeled as redacted, but images are forwarded to GPT-4o unmasked. This represents the class of text-only PII filtering approaches and serves as the primary comparison against prior-work-style methods. The key claim under test is that GPT-4o reads sensitive text directly from unmasked images, rendering text-level redaction ineffective.

Detective-Game Evaluation

Setup. The agent-level experiment frames privacy leakage as an identity-matching task. GPT-4o is given a set of evidence images and three fictional candidate profiles, and is asked to identify which candidate the evidence belongs to and to recover as many specific private attribute values as possible. The role-play framing avoids direct PII extraction requests, instead creating a naturalistic reasoning task in which the model may draw on any readable content within the images, cross-document correlations, or layout cues. Evidence images are assigned neutral labels ("Image 1," "Image 2," etc.); original filenames are not disclosed.

We compare three preprocessing conditions—*Raw*, *Text-Only*, and *Our Filter*—across all three individual profiles, yielding nine total runs.

Setting	Description
Dataset	Synthetic identity-document profiles constructed from Multi-Priv [2].
Individuals	Three fictional subjects: Amy (Illinois, USA); Anna Schmidt (Berlin, Germany); Li Na (Beijing, China).
Evidence per case	Up to ten document images per individual, across five document types.
Task	Match evidence to the correct candidate profile; recover specific attribute values.
Preprocessing modes	Raw, Text-Only, Our Filter.

Table 3: Setup of the detective-game evaluation.

Metrics. For each run, recovered attributes are scored against ground truth using exact matching for structured fields (ID numbers, card numbers, flight codes) and fuzzy matching (SequenceMatcher ratio ≥ 0.65) for free-text fields (names, addresses, diagnoses). We report:

Metric	Interpretation
Attribute recovery rate	$\mathrm{TP}/\lvert\mathrm{GT}\rvert$: fraction of ground-truth fields successfully extracted. Lower is better for privacy.
Recall	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$: fraction of true sensitive fields recovered. Primary privacy metric.
Precision	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$: fraction of claimed recoveries that were correct.
F1 score	Harmonic mean of precision and recall. Lower is better for privacy.
Identity accuracy	Whether the agent matched the evidence to the correct fictional individual.

Table 4: Metrics used in the detective-game evaluation.

Recall is the primary privacy metric: lower recall means less sensitive information was extractable from the preprocessed evidence.

Analysis and Results

Quantitative Results

Table 5 reports results aggregated over three cases per mode.

Mode	Attr. Recovery	Recall	Precision	F1	Identity Acc.
Raw	0.630	0.654	0.944	0.773	1.000
Text-Only	0.590	0.615	0.941	0.744	1.000
Our Filter	0.150	0.160	0.667	0.258	1.000

Table 5: Detective-game evaluation results (3 cases per mode).

The raw baseline confirms substantial leakage: GPT-4o recovers 63% of all sensitive attribute fields with an F1 of 0.773 when images are unprocessed.

The text-only baseline produces a marginal improvement: attribute recovery falls from 0.630 to 0.590 (−6%) and F1 from 0.773 to 0.744. This result directly demonstrates the inadequacy of text-level filtering—GPT-4o reads sensitive text from unmasked images using its built-in visual pathway, bypassing any redaction applied to the accompanying prompt.

Our full pipeline reduces attribute recovery to 0.150 and recall to 0.160, a **76% reduction** relative to the raw baseline:

$$\frac{0.630 - 0.150}{0.630} \approx 0.762$$

Examining individual cases, under our filter the agent in Case 1 recovered only the medical diagnosis field ("Normal abdominal ultrasound"); name, ID number, address, bank card number, and flight number all returned null. Diagnosis information partially persisted because medical report layout (the structural appearance of an imaging form) remains informative even after specific textual content is masked.

Identity Accuracy and Contextual Leakage

Identity accuracy remains at 1.000 across all three conditions. Under our filter, GPT-4o correctly matched each evidence set to its candidate profile using high-level behavioral cues—document category, travel direction, and approximate geographic region—rather than specific PII values. The candidate profiles were written in behavioral terms (e.g., "recently made a direct international flight between Frankfurt and Shanghai"), and this level of semantic information is preserved through masking.

This reveals a form of *contextual privacy leakage*: even when specific identifiers are suppressed, the combination of document type, travel pattern, and lifestyle category can uniquely identify an individual given sufficiently distinctive profiles. Eliminating this leakage would require full ABSTRACT suppression for all sensitive document categories, at significant utility cost.

The precision drop under our filter (0.944 to 0.667) reflects a small number of partially-correct inferences made from residual contextual cues. In Case 2, for example, the agent inferred "Berlin" as an approximate address from a Frankfurt departure city visible in layout, rather than the full street address on the masked ID card, scoring as a false positive against the exact ground truth.

Insights from the experiments

Visual masking is necessary. The near-identical performance of the raw and text-only conditions confirms that text redaction at the prompt level provides no effective protection when images are forwarded unmasked to a vision-capable model. Any privacy system that operates solely on textual input is trivially bypassed by a model that perceives document content through its visual pathway.

MASK vs. ABSTRACT. Under the MASK policy, specific field values are suppressed, but document structure (layout, visual format, spatial relationships) remains visible. Our results show that an LLM can still partially reason about document type and approximate content from masked images. The ABSTRACT policy eliminates this structural leakage by dropping the image entirely, but reduces visual utility. The appropriate choice between MASK and ABSTRACT involves a direct privacy-utility tradeoff and depends on the sensitivity classification of the document type.

Unimplemented pipeline stages. The current system gates privacy at the sensing and memory-write stages only. The retrieval stage—which assembles memory candidates from the vector store and injects them into agent context—and the output stage—which should verify the agent's final response for leaked content—were not implemented as controlled evaluation conditions. These represent known gaps where sensitive memories could re-enter the reasoning pipeline even after input-side filtering.

Limitations

Limitation	Description
Small evaluation scale	Due to token limitations, the detective-game evaluation covers three fictional individuals (nine total agent runs). Results may not generalize to broader document types, subject populations, or adversarial query strategies.
Contextual leakage	Identity matching accuracy remained at 100% across all conditions, showing that behavioral patterns (document type, travel direction, approximate location) are sufficient for re-identification even when all specific identifiers are masked.
Document-structure leakage	The MASK policy preserves image layout. An adversary with knowledge of document formats can infer document category from masked images, which itself carries privacy-relevant information.
Single-attribute re-identification	Exposure of one highly specific attribute (e.g., an unusual flight route) may enable re-identification in a small candidate pool, even when all other fields are suppressed.
No retrieval or output gate	The retrieval and output stages of the agent pipeline are unguarded. Private content that persists in the vector store may re-enter agent context at query time, and leaked content in final responses is not checked.
Runtime overhead	Running four detection models locally (NER, OWL-ViT, Haar cascade, EasyOCR) introduces multi-second latency per memory event, which may be prohibitive for real-time agent settings.
Synthetic evaluation data	The evaluation uses synthetically constructed identity documents. Generalization to real-world document noise, diverse layouts, and non-English content has not been tested.

Table 6: Limitations of the current system.

Future Work

Retrieval and output gates. Addressing the missing pipeline stages (Section 6) requires implementing privacy filtering at both the retrieval and output stages. A retrieval gate would re-evaluate each vector candidate against the current query before injecting it into agent context, preventing

sensitive memories from re-entering the reasoning pipeline. An output gate would scan the agent's final response for leaked identifiers or raw object references before they are returned to the user.

Structured entity abstraction. The MASK policy replaces sensitive regions with solid black rectangles, which discards visual content entirely. Replacing opaque black fills with semantically consistent placeholders—for example, rendering a blurred or noise-filled region at the same spatial location—would preserve more document structure while suppressing specific values, reducing the document-structure leakage identified in our results.

Broader evaluation. Addressing the small-scale evaluation limitation requires a larger dataset spanning more individuals, document types, image qualities, and non-English content. An adversarial evaluation protocol—where the downstream model is specifically prompted to probe for remaining leakage paths—would more rigorously test pipeline robustness than the cooperative detective-game framing used here.

Contextual and behavioral privacy. The identity accuracy result reveals a limitation that detection and masking alone cannot fully address: behavioral patterns composed from multiple individually innocuous signals can be as identifying as direct PII. Future work should explore privacy-preserving memory construction—for example, introducing controlled generalization into stored behavioral facts—to reduce the fingerprinting surface left by high-level contextual cues.

Domain-specific detection. The current OWL-ViT query vocabulary was defined manually and is fixed at inference time. Fine-tuning a visual detector on a labeled document privacy dataset, or learning a vocabulary-expansion mechanism, would improve recall for specialized document types without increasing the false-positive rate.

Conclusion

We presented a privacy-gated multimodal agent memory system that intercepts sensitive content in both text and image channels before CLIP embedding. The system combines NER-based text redaction, zero-shot visual object detection, deterministic face detection, and ephemeral OCR into a unified sensing gate, and applies a three-tier policy (ALLOW / MASK / ABSTRACT) determined by the fraction of detected sensitive image area.

A controlled detective-game evaluation demonstrates that our full pipeline reduces sensitive attribute recovery by 76% relative to an unguarded baseline (attribute recovery: $0.630 \rightarrow 0.150$; recall: $0.654 \rightarrow 0.160$), while preserving identity-level task accuracy at 100%. Critically, text-only filtering reduces attribute recovery by only 6%, confirming that visual masking is a necessary—not optional—component of multimodal privacy protection. When document images are forwarded to a vision-capable model unmasked, text-level redaction provides no effective barrier.

These results support the central claim: effective privacy in multimodal agent memory systems requires that privacy enforcement happen at the input level, before content is embedded, and that it operate over both text and image modalities jointly. Remaining gaps—particularly at the retrieval and output stages, and for contextual leakage through behavioral patterns—represent concrete directions for future work.

We expect this work to be useful in fields where privacy-protection is the priority, e.g. avoid leakage of patients' privacy in using LLM for medical diagnosis or avoid leakage of clients' privacy in using LLM for legal or financial documents.

Ethic Statement

The dataset we used is a synthetic dataset.

References

[1] Microsoft. *Presidio: Data Protection and De-identification SDK*. `https://github.com/microsoft/presidio`, 2019.

[2] CyberChangAn. *MultiPriv: A Synthetic Multimodal PII Dataset*. `https://github.com/CyberChangAn/MultiPriv`, 2024.